

Global Semiconductor Market Size

\$627 billion in 2024

\$1 trillion in 2030 (estimated)

\$2 trillion by 2040 (estimated)

SOURCE DELOITTE

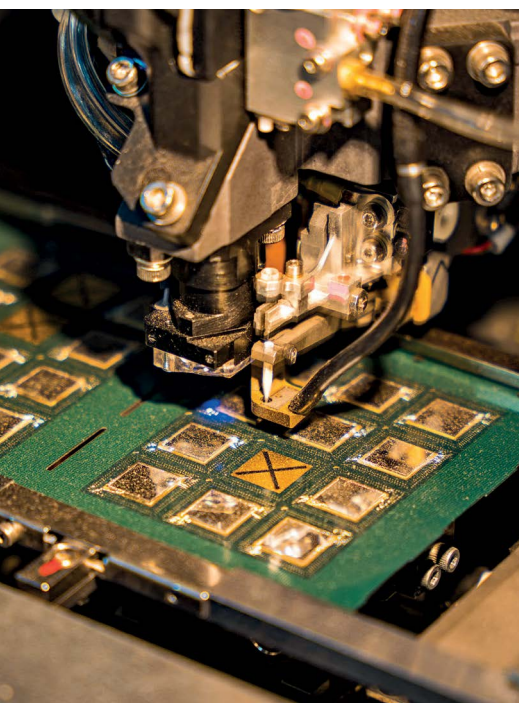


PHOTO BY AFP

India Semiconductor Market Size

\$45-50 billion in 2024-25

\$100-110 billion in 2030 (estimated)

SOURCE MEITY

crore semiconductor mission in December 2021. With a vast electronics market and a robust design-services sector, it can become a serious player in the global semiconductor ecosystem, say experts.

India's initial breakthroughs, though, have come on the back-end side. A significant moment was US memory giant Micron committing ₹22,516 crore for an ATMP plant in Sanand. This spurred domestic conglomerates. Murugappa's CG Power tied up with Renesas (Japan) and Stars Microelectronics (Thailand) for a ₹7,600 crore OSAT facility. "We must start thinking of the national need. It is critical for us," says Vellayan Subbiah, Chairman, CG Power. Explaining the decision to start with packaging, he says, "It is too early for fabs. But (we) have already announced a foray into design and acquired a company (radio frequency components business of Renesas) to enter design. If you think of the profit pool, a significant value comes from fabless design. So, we (have) started with that, other than OSAT." Fabless refers to a business model where a company outsources manufacturing and focuses on design, development, and marketing of semiconductor chips. CG Semi's facility will be able to package around 15 million chips per day once fully scaled up.

It's not alone. HCL has partnered with Foxconn for an OSAT plant in Jewar in Uttar Pradesh. The government recently cleared four more projects, including CDIL's Mohali expansion. "With our indigenously developed semiconductor packaging technology at its core, our Mohali expansion will add capacity in power semiconductors," says Prithvideep Singh, General Manager, CDIL.

INDIA'S PUSH

India's chip story began with ambition in 1984 when it set up Semiconductor Complex Ltd in Mohali. The company moved quickly from 5,000nm to 800nm chips. It was just a step behind global leaders and ahead of China and Taiwan. A devastating fire at the plant in 1989 crashed India's semiconductor dreams. It reopened in 1997 but now survives as Semi-Conductor Laboratory. Several big proposals over the next few years—including Intel's \$2.5 billion plan in 2004-05—did not see the light of day, not the least due to government indifference.

Is it different this time? In a sense, yes. Under the ISM, the Centre has committed to covering 50% costs of fab, testing and packaging plants. Several state governments have announced an additional 20-25% support. Yet, leading players such as TSMC, Samsung, GlobalFoundries, and Intel continue to direct multi-billion-dollar investments into the US, Japan, and Europe. The reasons include low customer base, a perceived mismatch between risk and return, and a still-developing support ecosystem.

TSMC, the world's leading foundry, says its expansion is based on an interplay of factors, including customer needs, business opportunities, operating efficiency, government support, and cost and economic considerations. "At present, our overseas expansion is focused on our previously announced projects in Japan, Germany and the US. While we do not rule out future possibilities, we do not currently have plans to explore other options," says a TSMC spokesperson.

The company announced its US fab in May 2020 to serve clients such as Apple, followed by its Japan fab in November 2021, both before